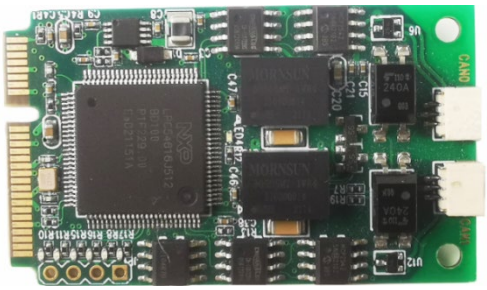
SavvyCAN-FD Series Products

Pibiger High Speed USB 2.0 To CAN FD Converter

Data Field Up to 12M Max/ CUSTOM Nominal/Data Bit Rate

USB To CANFD Series Products

User Manual

		
SavvyCAN-FD-X2 USB To Dual Channel CANFD	SavvyCAN-FD-C USB To CANFD Cable	SavvyCAN-FD-M USB To Dual Channel CANFD with mini pcie interface (usb2.0 bus)

Date	Revision	Description
2023/01/01	V1.0	First Released
2025/09/11	V1.1	Add Software User Guide Chapter3,4,5

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1INTRODUCTION

1.1 Product Series

Product Name	CAN Version	Bitrate	Isolation	Max Rate	Interface	Time Stamp
SavvyCAN-FD-X2	CAN2.0A/B CANFD 1.0	12Mbit/s	2500V	15000fps/s	USB 2*DSUB 9PIN	1us
SavvyCAN-FD-mPCI	CAN2.0A/B CANFD 1.0	12Mbit/s	2500V	15000fps/s	MINIPCIE (USB Bus)	1us
SavvyCAN-FD-C	CAN2.0A/B CANFD 1.0	12Mbit/s	2500V	15000fps/s	1.5M USB Cable 1*D-Sub 9PIN	1us

The new CAN FD standard (CAN with Flexible Data rate) is primarily characterized by higher bandwidth for data transfer.

The maximum of 64 data bytes per CAN FD frame (instead of 8 so far) can be transmitted with bit rates up to 12 Mbit/s.

1.2 Product Features

1 True High-speed USB 2.0 Compatible with CAN specifications 2.0A/2.0B/FD;

2 CAN FD support for ISO and Non-ISO standard support software switch;

3 CAN FD bit rates data field (64 bytes max.) from 25 kbit/s up to 12Mbit/s;

4 Class CAN bit rates data field from 25 kbit/s up to 1 Mbit/s;

5 Time stamp Resolution Up to 1 μs;

6 Each CAN FD Signal &Power Separately Isolated Up to 2500 Volts against USB;

7 Bus load measurements including error frames and overload frames on physical bus;

8 Induced error generation for incoming and outgoing CAN messages;

2HARDWARE DESCRIPTION

2.1 SavvyCAN-FD-X2

The SavvyCAN-FD-X2 is a plug and play high speed USB2.0 to CANFD adapter enables the connection of dual channel CANFD networks to a computer via USB. Each CAN FD channel is separately isolated against USB with a maximum of 2500V.

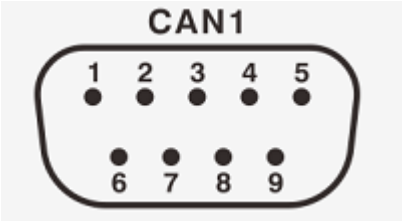
Specification

Connector	
CANFD	Dual Channel 9PIN D-SUB Connectors
USB	USB plug type A（Computer） USB plug type B（Devcie）
CAN Features	
Protocols	CAN 2.0A (standard format) CAN 2.0B (extended format) CAN FD ISO 11898-1:2015 CAN FD non-ISO
CAN bit rates	25 kbit/s up to 1 Mbit/s
CANFD bit rates	25 kbit/s up to 12Mbit/s
USB ESD	IEC 61000-4-2 Level 4 +25 kV (contact discharge) +30 kV (air-gap discharge)
Galvanic isolation	Signal &Power Separately Isolated by 2500 Volts against USB IEC 61000-4-2 Contact IEC 61000-4-2 Air ISO 10605 150 pF / 2 k2 Contact ISO 10605 330 pF 1 2 k2 Contact
Micro controller	180MHZ Cortex-M4 MCU
Timestamp resolution	1 μs

Built In 120Ω Termination Resistor	Enable/Disable Through Software
Others	
Temperature	-40°~ 85°
PCBA Size (L * W * H)	84x80x28 mm
Weight	190 g

Pins Out Description

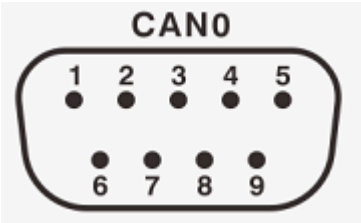
CAN1



PIN OUT Description

1	NC
2	CANL bus line (dominant low)
3	CAN_GND
4	NC
5	NC
6	NC
7	CANH bus line (dominant high)
8	NC
9	NC

CAN0（Normal）



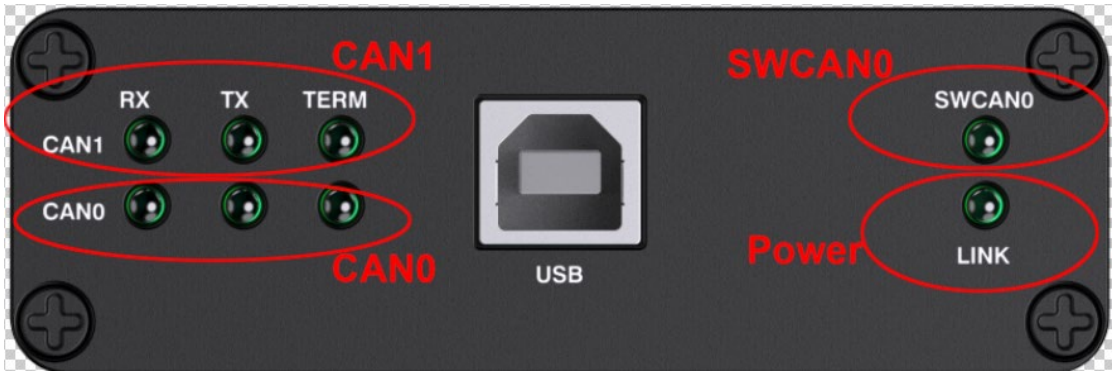
PIN OUT Description

1	NC
2	CANL bus line (dominant low)
3	CAN_GND

	4	NC
	5	NC
	6	NC
	7	CANH bus line (dominant high)
	8	NC
	9	NC

Note1: Not connect GND do not affect normal communication, if cable with shielding suggest connect to GND;

LED Indication



When plug SavvyCAN-FD-X2 device to Computer All lights are flashing for one second.

Then TERM LED And LINK LED turns to be green.

Led Name	Description
Power LED Indicator	Power Up And Driver Installted LINK LED turns to be blinking.
CAN1 LED Indicator	TX LED Blinking, Sending Data;

	RX LED Blinking, Receiving Data; TERM LED Green,120Ω Activated;
CAN0 LED Indicator	TX LED Blinking, Sending Data; RX LED Blinking, Receiving Data; TERM LED Green,120Ω Activated;

2.2 SAVVYCAN-FD-C

The SAVVYCAN-FD-C is a plug and play high speed USB2.0 to CANFD converter comes with 1.2m high quality Cable enables the connection of one channel CANFD networks to a computer via USB. CANFD is isolated protection against USB 2500V.

SavvyCAN-FD Series Products

Pibiger High Speed USB 2.0 To CAN FD Converter

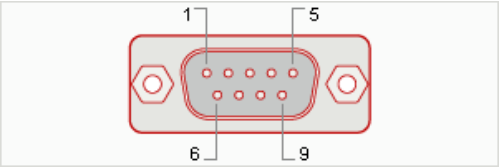
Data Field Up to 12M Max/ CUSTOM Nominal/Data Bit Rate

Specification

Connector	
CANFD	9PIN D-SUB Connectors
USB Cable	1.2M High Quality USB Cable with type A
CAN Features	
Protocols	CAN 2.0A (standard format) CAN 2.0B (extended format) CAN FD ISO 11898-1:2015 CAN FD non-ISO
CAN bit rates	25 kbit/s up to 1 Mbit/s
CANFD bit rates	25 kbit/s up to 12Mbit/s
USB ESD	IEC 61000-4-2 Level 4 +25 kV (contact discharge) +30 kV (air-gap discharge)
Galvanic isolation	Signal &Power Separately Isolated by 2500 Volts against USB IEC 61000-4-2 Contact IEC 61000-4-2 Air ISO 10605 150 pF / 2 k2 Contact ISO 10605 330 pF 1 2 k2 Contact
Micro controller	180MHZ Cortex-M4 MCU
Timestamp resolution	1 μs
Built In 120Ω Termination Resistor	Disable/Enable Through Software

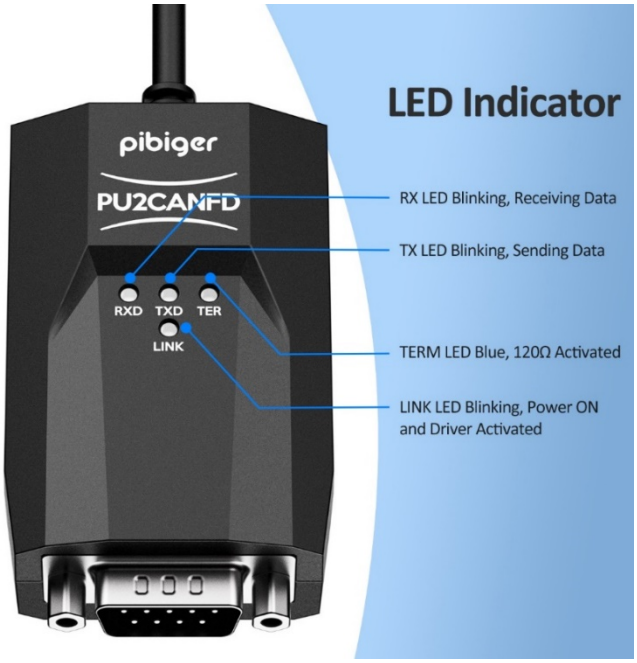
Others	
Temperature	-40°~ 85°

Pins Out Description



1	NC
2	CAN-L
3	GND
4	NC
5	NC
6	NC
7	CAN-H
8	NC
9	NC

LED Indication



Led Name	LED Status	Description
RXD	Blinking	Receiving Data
TXD	Blinking	Sending Data
TER	Blue Led On	Build In 120Ω Term Resistor Enable;
LINK	Blinking	Power Up and Driver Installed.
	Blue Led On	Power Up Driver Not Installed.

2.3 SavvyCAN-FD-M

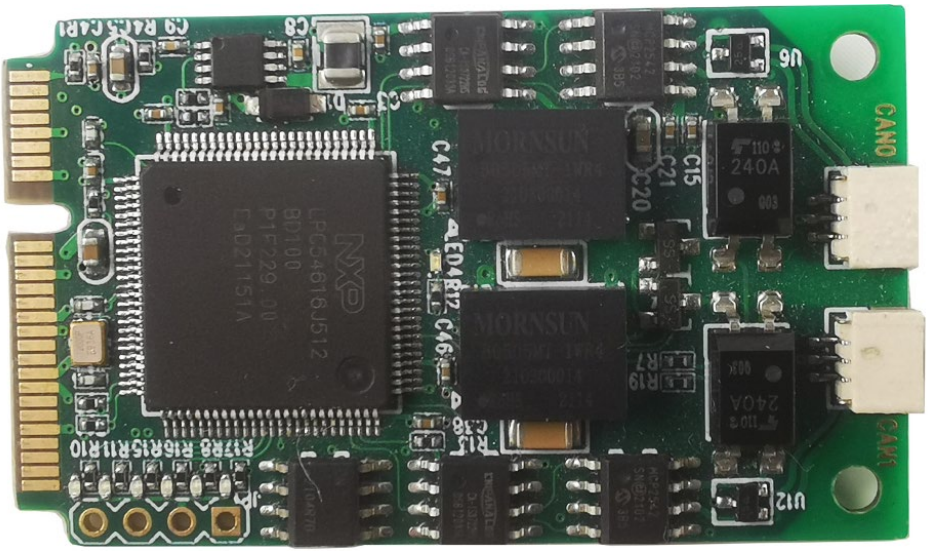
The SavvyCAN-FD-M is a plug and play high speed USB2.0 to CANFD CAN Card.

CAN FD each channel is separately isolated against USB with a maximum of 2500V.

SavvyCAN-FD Series Products

Pibiger High Speed USB 2.0 To CAN FD Converter

Data Field Up to 12M Max/ CUSTOM Nominal/Data Bit Rate



Specification

Connector	
Dual Channel CANFD	2X3PIN 1mm Connector
Mini PCIE USB	USB BUS
CAN Features	
Protocols	CAN 2.0A (standard format) CAN 2.0B (extended format) CAN FD ISO 11898-1:2015 CAN FD non-ISO
CAN bit rates	25 kbit/s up to 1 Mbit/s
CANFD bit rates	25 kbit/s up to 12Mbit/s
USB ESD	IEC 61000-4-2 Level 4 +25 kV (contact discharge) +30 kV (air-gap discharge)

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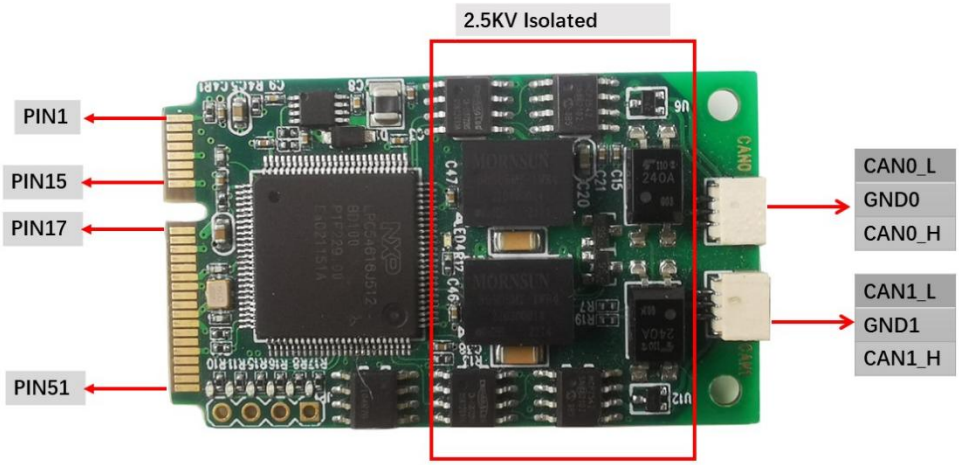
Galvanic isolation	Signal &Power Separately Isolated by 2500 Volts against USB IEC 61000-4-2 Contact IEC 61000-4-2 Air ISO 10605 150 pF / 2 k2 Contact ISO 10605 330 pF 1 2 k2 Contact
Micro controller	180MHZ Cortex-M4 MCU
Timestamp resolution	1 μs
Built In 120Ω Termination Resistor	Activated/Deactivated Through Software
Others	
Temperature	-40°~ 85°
PCBA Size (L * W * H)	84x80x28 mm
Weight	191 g

Pins Out Description



SavvyCAN-FD Series Products

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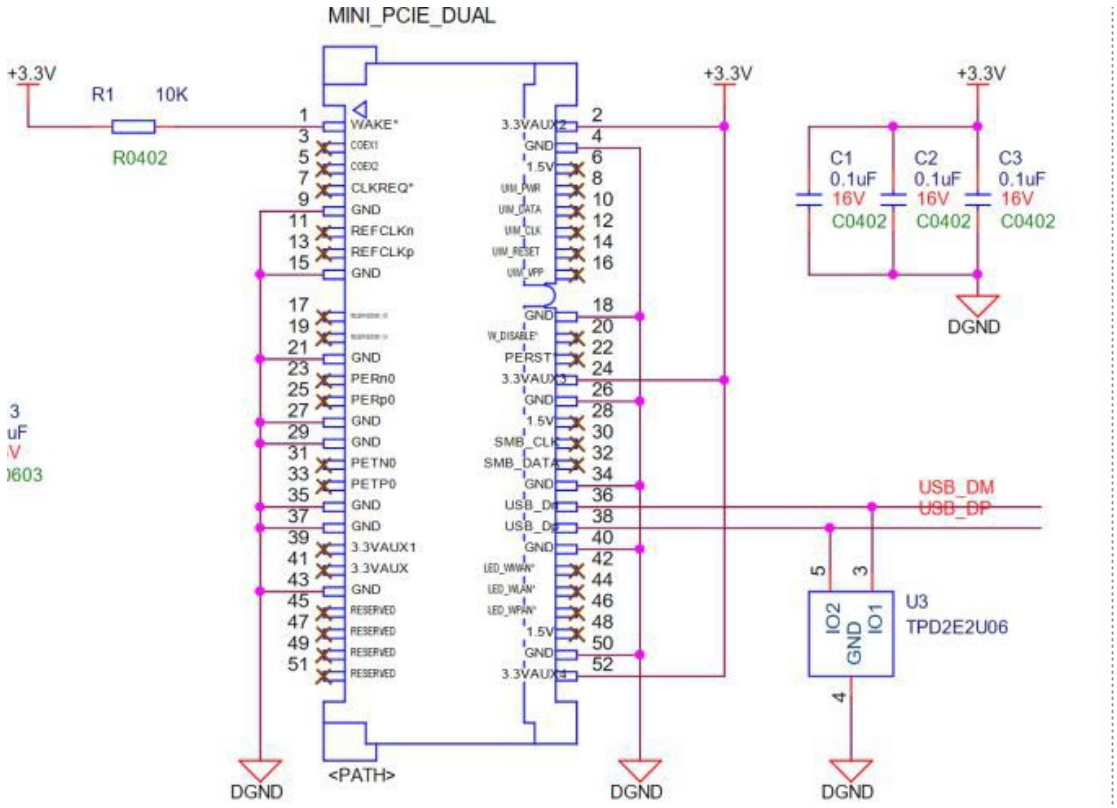
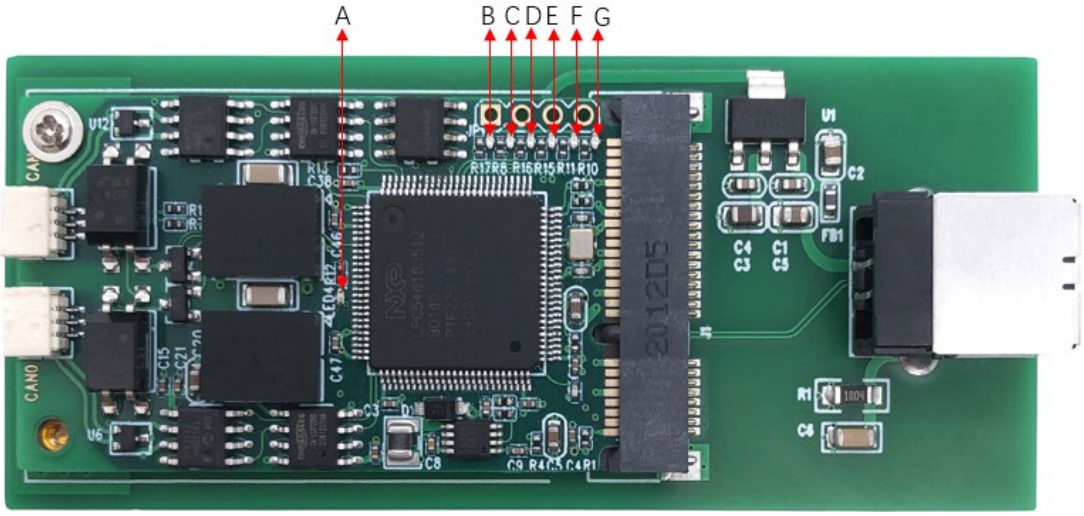
Data Field Up to 12M Max/ CUSTOM Nominal/Data Bit Rate



LED Indication



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	Description	Indication
A	Power&Driver LED	Blinking, Driver install and Power Up
B	CAN0 120Ω Term	On, CAN0 Term Enable; Off, CAN0 Term Disable
C	CAN1 120Ω Term	On, CAN1 Term Enable; Off, CAN1 Term Disable
D	CAN0 Receive	Blinking，CAN0 Receiving Data;
E	CAN0 Send	Blinking，CAN0 Sending Data;
F	CAN1 Receive	Blinking，CAN1 Receiving Data;
G	CAN1 Send	Blinking，CAN0 Receiving Data;

Design Reference

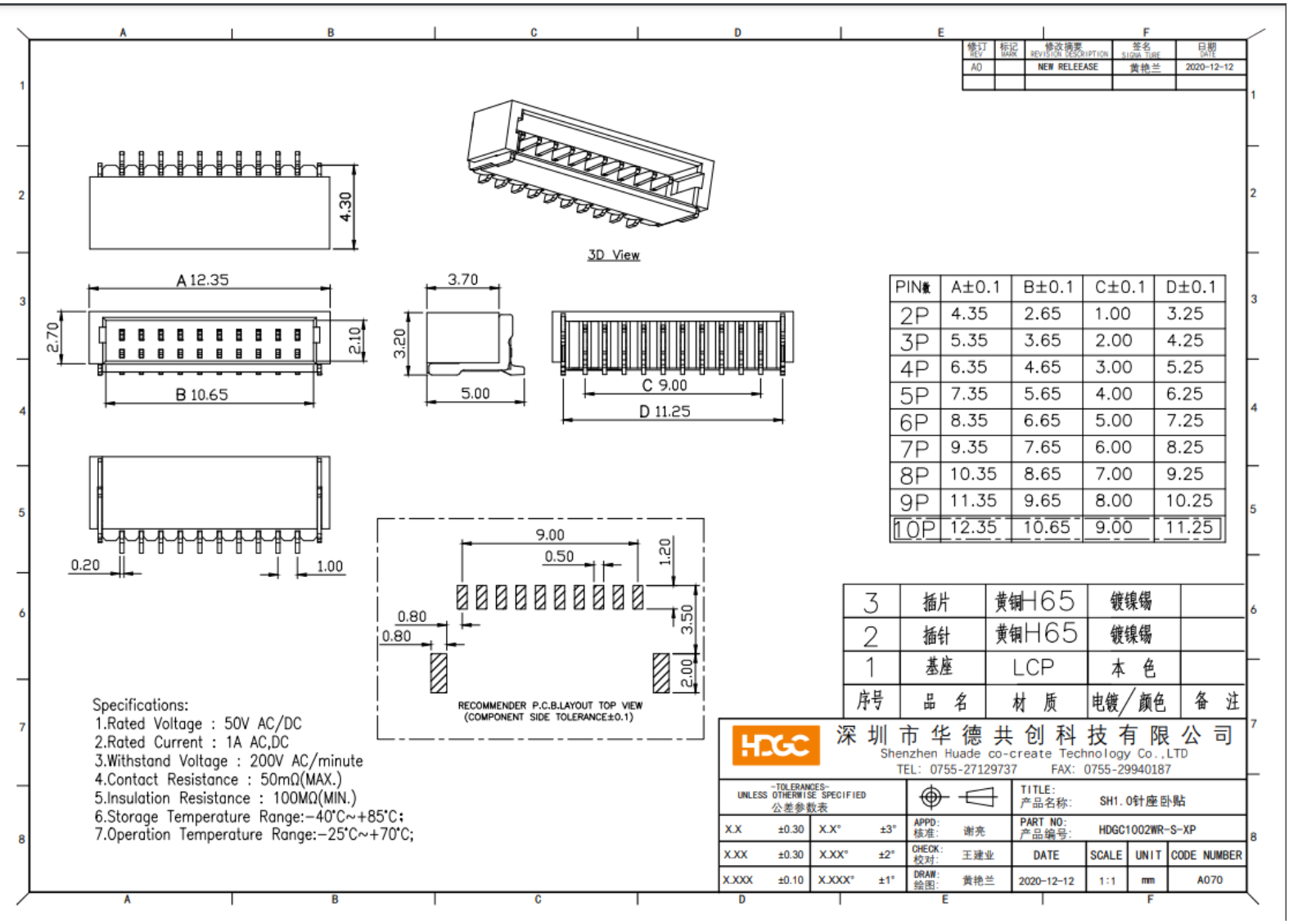
3PIN Connector Board End

Vendor：HDGC1002WR-S-3P

SavvyCAN-FD Series Products

Pibiger High Speed USB 2.0 To CAN FD Converter

Data Field Up to 12M Max/ CUSTOM Nominal/Data Bit Rate



PIN数	A±0.1	B±0.1	C±0.1	D±0.1
2P	4.35	2.65	1.00	3.25
3P	5.35	3.65	2.00	4.25
4P	6.35	4.65	3.00	5.25
5P	7.35	5.65	4.00	6.25
6P	8.35	6.65	5.00	7.25
7P	9.35	7.65	6.00	8.25
8P	10.35	8.65	7.00	9.25
9P	11.35	9.65	8.00	10.25
10P	12.35	10.65	9.00	11.25

序号	品 名	材 质	电镀/颜色	备 注
3	插片	黄铜H65	镀镍锡	
2	插针	黄铜H65	镀镍锡	
1	基座	LCP	本 色	

Design End

Vendor: HDGC1002H-5P

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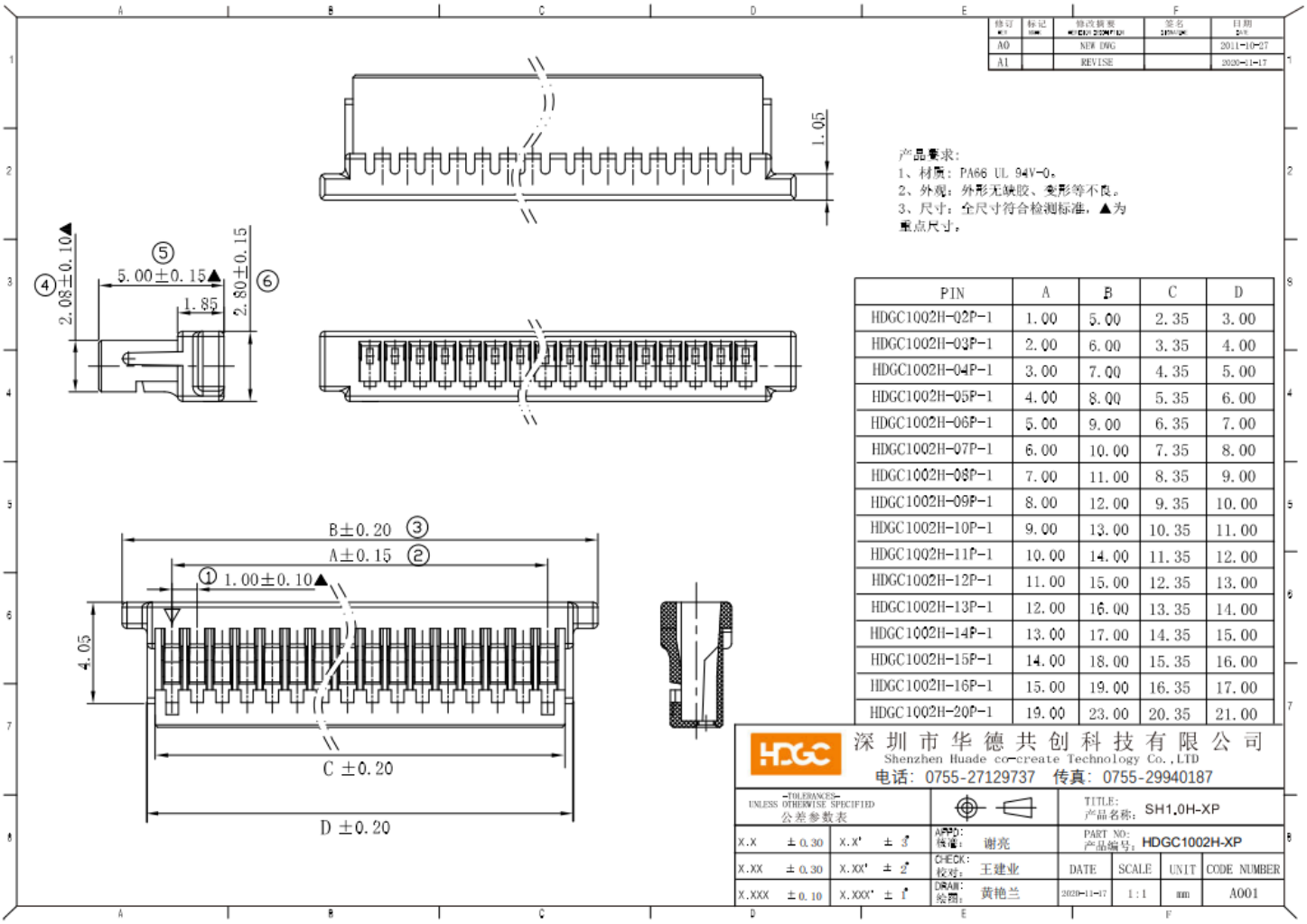
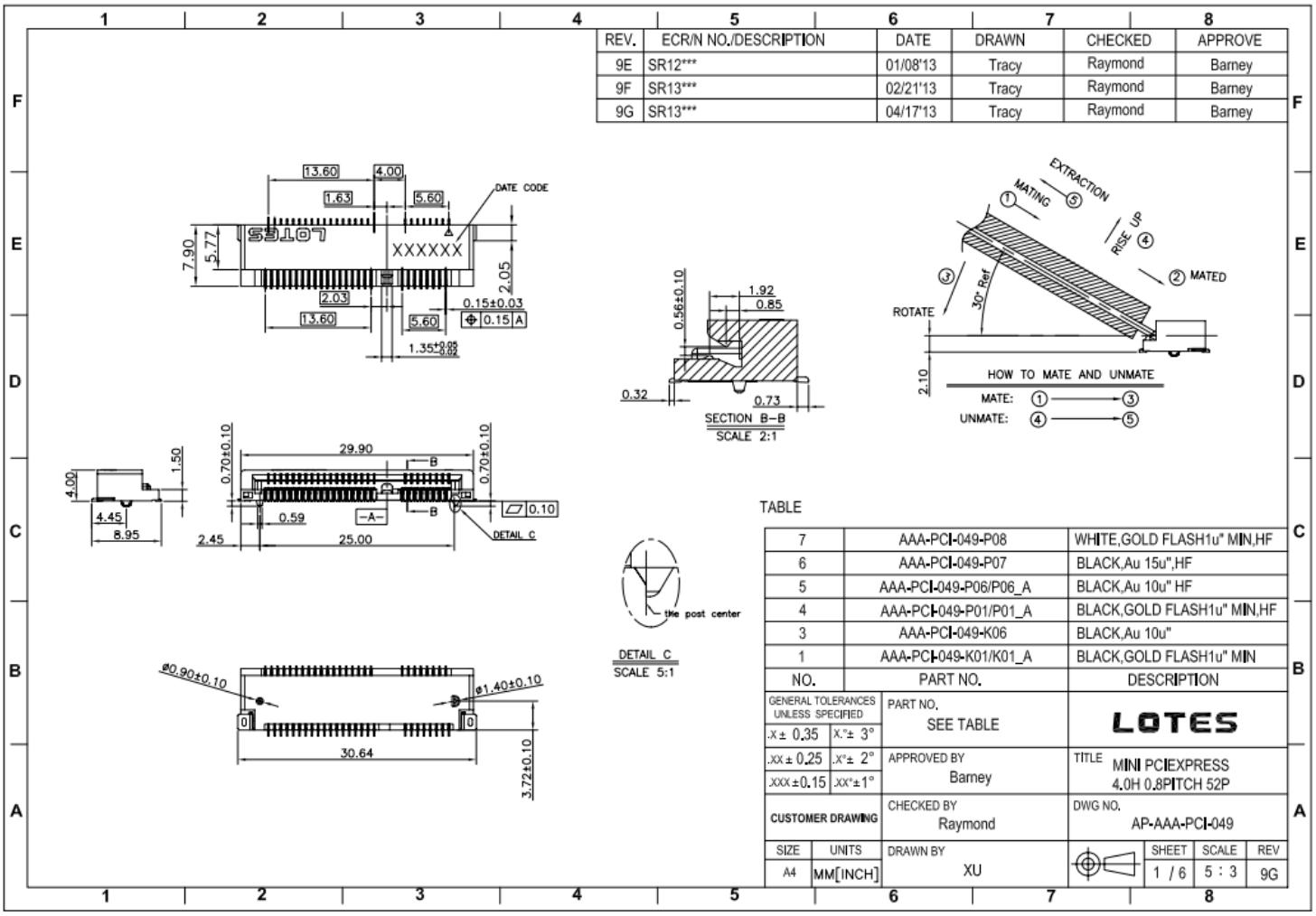


PIN	A	B	C	D
HDGC1002H-02P-1	1.00	5.00	2.35	3.00
HDGC1002H-03P-1	2.00	6.00	3.35	4.00
HDGC1002H-04P-1	3.00	7.00	4.35	5.00
HDGC1002H-05P-1	4.00	8.00	5.35	6.00
HDGC1002H-06P-1	5.00	9.00	6.35	7.00
HDGC1002H-07P-1	6.00	10.00	7.35	8.00
HDGC1002H-08P-1	7.00	11.00	8.35	9.00
HDGC1002H-09P-1	8.00	12.00	9.35	10.00
HDGC1002H-10P-1	9.00	13.00	10.35	11.00
HDGC1002H-11P-1	10.00	14.00	11.35	12.00
HDGC1002H-12P-1	11.00	15.00	12.35	13.00
HDGC1002H-13P-1	12.00	16.00	13.35	14.00
HDGC1002H-14P-1	13.00	17.00	14.35	15.00
HDGC1002H-15P-1	14.00	18.00	15.35	16.00
HDGC1002H-16P-1	15.00	19.00	16.35	17.00
HDGC1002H-20P-1	19.00	23.00	20.35	21.00

MINIPCIE Connector

Vendor: Lotes, AAA-PCI-049-K01



3 CONNECTION

3.1 Enable/Disable Build In 120Ω Term Resistor

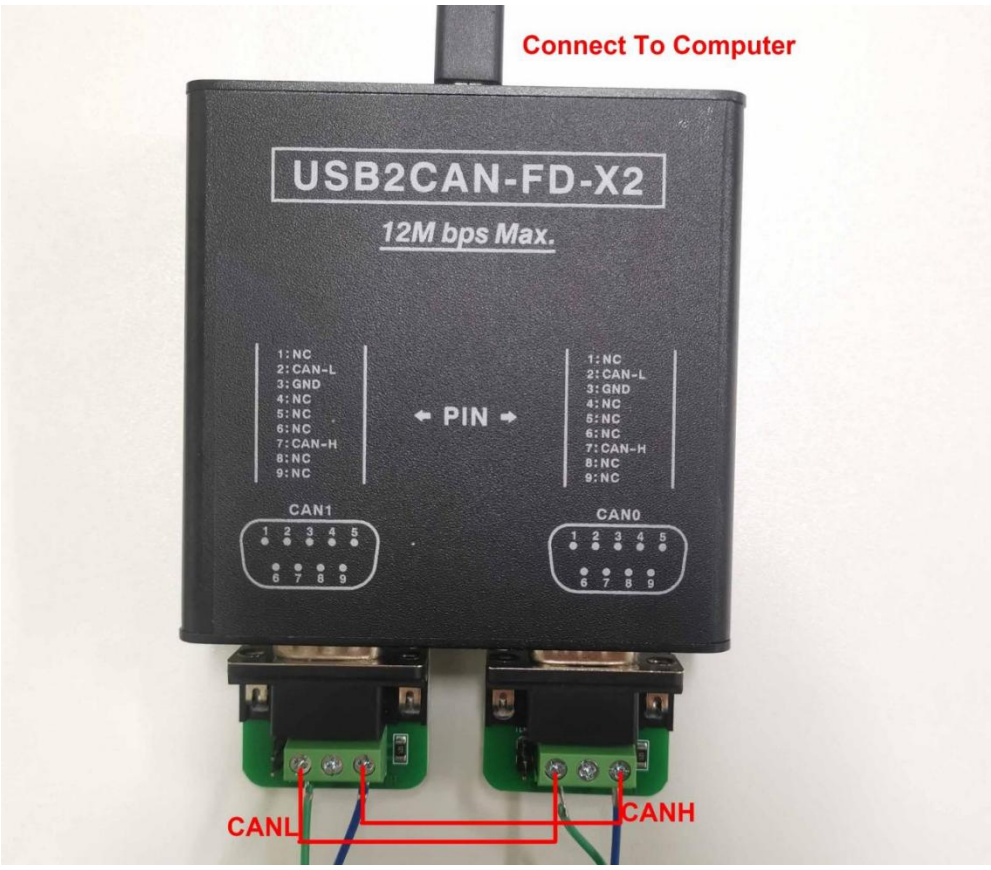
Products **comes enable build in 120Ω term resistor in default**.

If you need external connection of 120Ω termination resistor please refer to below method:

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/quick-start-guide#id-1-5-internal-120o-termination-resistor

3.2 SavvyCAN-FD-X2/SAVVYCAN-FD-C Connection

CAN 0 Channel	Connection	CAN 1 Channel
CAN_L(pin 2)	----------	CAN_L(pin 2)
CAN_H(pin 7)	----------	CAN_H(pin 7)



Notes:

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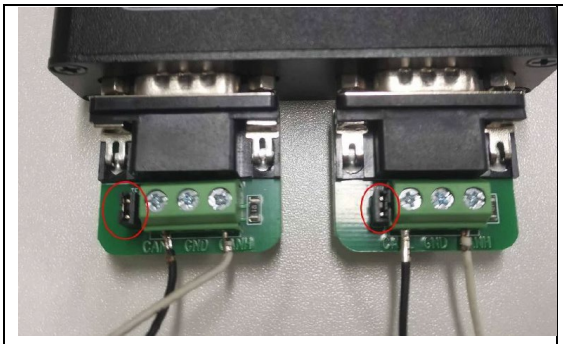
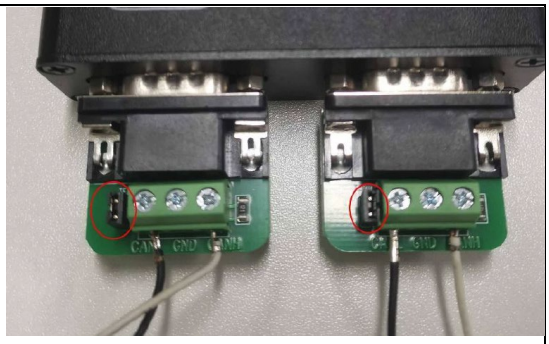
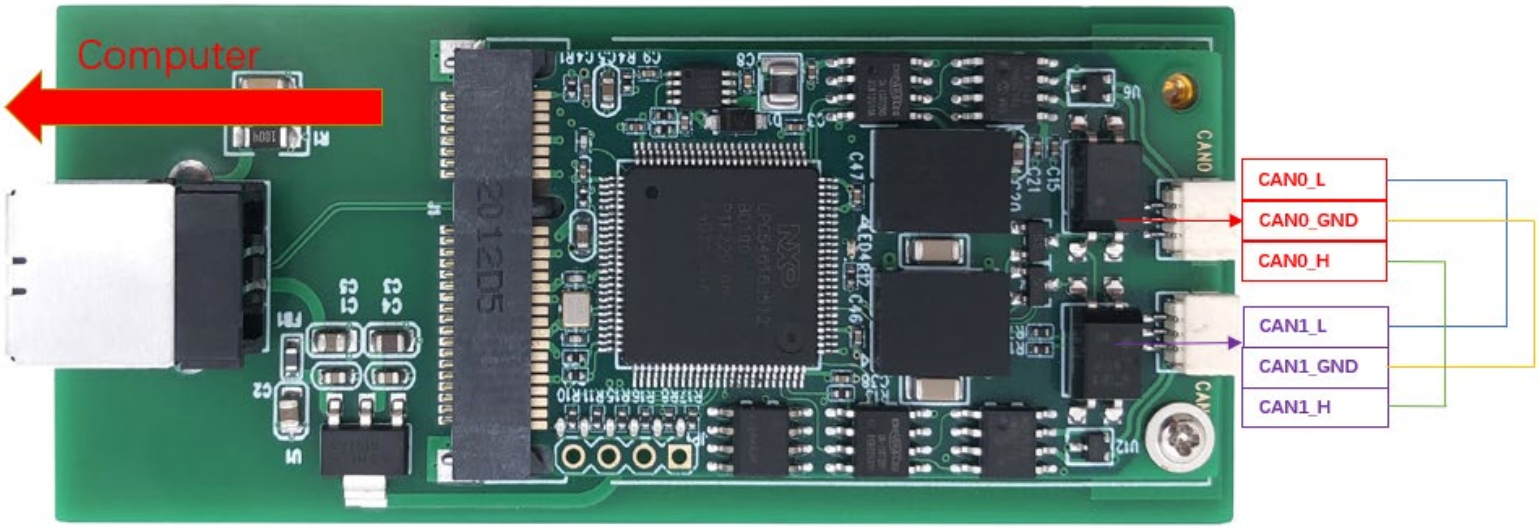
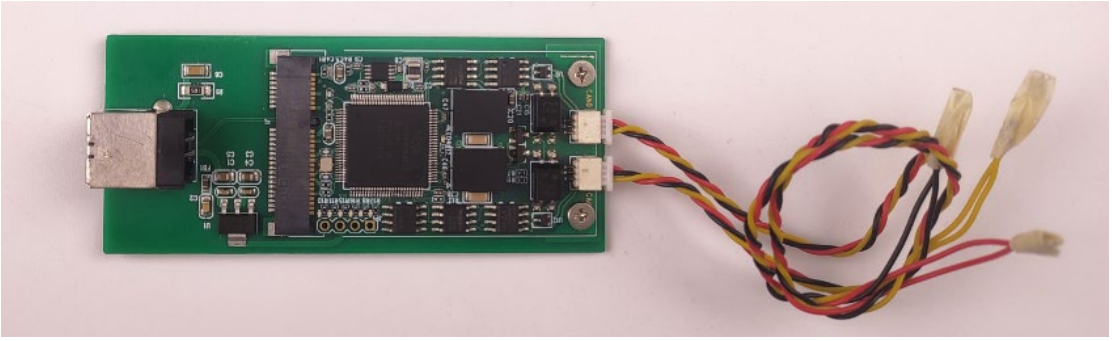
	
Note: Remove jumper When build in 120Ω termination resistor Enable.	**Note: Add jumper When build in 120Ω termination resistor Disable.**

3.3 SavvyCAN-FD-mPCI Connection Figure





4 DEVICE OPERATION

Note： USB To CANFD Series Products comes

- Enable 120Ω termination resistor,
- Default working on windows OS.

If you need to change to Linux OS/Mac OS/ Disable 120Ω termination resistor, you need to change device ID.

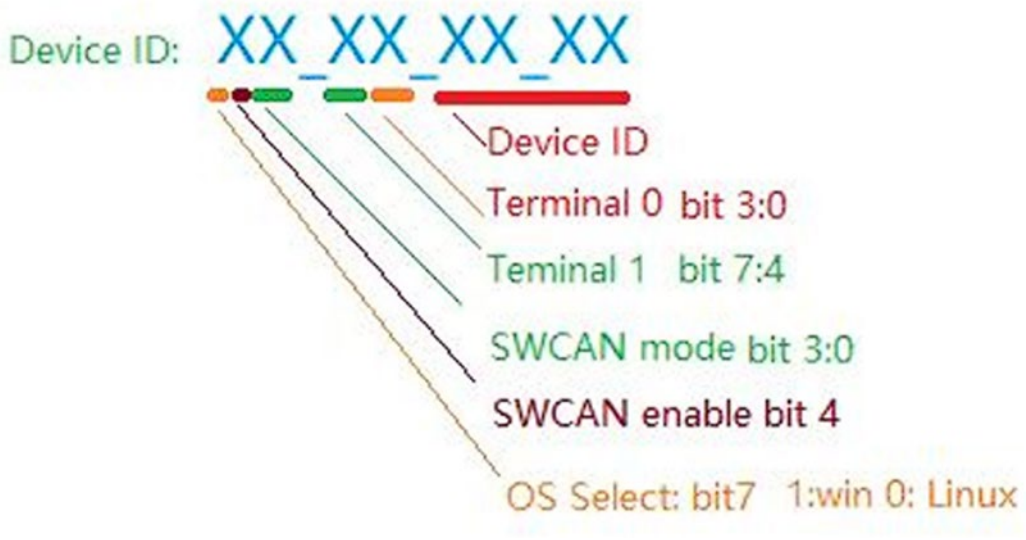
4.1 Device ID Description

Device serial number have 4 bytes: Byte3|Byte2|Byte1|Byte0

Device serial number byte 2 used to control 2 channels of terminal resister.

BYTE2 bit0—3 control channel0--- 1: resister on 0 :terminal resister off

BYTE2 bit7—4 control channel1--- 1: resister on 0 :terminal resister off



4.2 Device ID Change Method

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/quick-start-guide#id-1-7-change-product-id-for-difference-os

Note: After Setting ID, reconnect USB Cable to PC then take effect.

4.3 Internal 120Ω Term Resistor Enable/Disable

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/quick-start-guide#id-1-5-internal-120o-termination-resistor

4.4 CUSTOM Nominal/Data Bit Rate

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/quick-start-guide#id-1-6-custom-nominal-data-bit-rate

4.5 CAN FD Protocol

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/can-fd-protocol

5 SOFTWARE

5.1 Software Description

This device Support Windws,Linux,Mac Os.

Windows:

- Pibiger SavvyCAN-FD-Version On Windows

Linux:

- SavvyCAN Open Source Software

MacOs

- SavvyCAN Open Source Software

Note: Pibiger Default Provide Software savvyCAN with

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FD Support. You can download from below link:

SavvyCAN-FD-Version

5.2 SavvyCAN User Guide On Windows

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/quick-start-guide#id-1-2-how-to-use-on-windows

5.3 SavvyCAN User Guide On Linux

Follow Chapter 4.2 Change Device ID For Linux First

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/quick-start-guide#id-1-3-how-to-use-on-linux

5.4 SavvyCAN User Guide On Mac Os

Follow Chapter 4.2 Change Device ID For Linux First

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/quick-start-guide#id-1-4-how-to-use-on-mac-os

5.5 Third Party Software

Follow Chapter 4.2 Change Device ID For Windows/Linux/MacOS First

https://docs.pibiger-tech.com/home/usb-to-can-fd-series/savvycanfd/third-party-software-support

5.6 Open Source CAN-UTILS/C/Python For Linux

（USE AS SOCKET CAN）

5.6.1 Linux Support List

USB2CANFD-X2 device can run properly without any additional driver request on all Linux system as below.

	amd64	i386	arm64	armhf	ppc64el
Ubuntu:					
Trusty 14.04 LTS	■	■	■	■	
Xenial 16.04 LTS	■	■	■	■	■
Bionic 18.04 LTS	■	■	■	■	■
Cosmic 18.10	■	■	■	■	
Disco 19.04	■	■	■	■	
Eoan 19.10	■	■	■	■	
Focal 20.04 LTS	■	■	■	■	■
Groovy 20.10	■	■	■	■	■
Hirsute 21.04	■	■	■	■	■
OpenSUSE Tumbleweed	see Xenial				
Debian:					
Wheezy 7.11	■	■	■	■	
Jessie 8.11	■	■	■	■	

Stretch 9.9	■	■	■	■	
Buster 10	■	■	■	■	■
Bullseye 11	■	■	■	■	■

5.6.2 CAN-UTILS DEMO

Prepare

Type command ifconfig -a to check 'can0' and 'can1'device is available in system, if you can not find the command ifconfig, use command sudo apt-get install net-tools

```
virtual-machine:~$ ifconfig -a
can0: flags=128<NOARP>  mtu 16
        unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00  txqueuelen 10  (UNSPEC)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

can1: flags=128<NOARP>  mtu 16
        unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00  txqueuelen 10  (UNSPEC)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Type command dmesg to check more information

```
[ 4334.851804] usb 1-1: new high-speed USB device number 6 using ehci-pci
[ 4335.120375] usb 1-1: New USB device found, idVendor=0c72, idProduct=0011, bcdDevice= 0.00
[ 4335.120380] usb 1-1: New USB device strings: Mfr=1, Product=2, SerialNumber=0
[ 4335.120384] usb 1-1: Product: PCAN-USB Pro FD
[ 4335.120385] usb 1-1: Manufacturer: PEAK-System Technik GmbH
[ 4335.137143] peak_usb 1-1:1.0: PEAK-System PCAN-USB Pro FD v2 fw v3.2.0 (2 channels)
[ 4335.143995] peak_usb 1-1:1.0 can0: attached to PCAN-USB Pro FD channel 0 (device 0)
[ 4335.151388] peak_usb 1-1:1.0 can1: attached to PCAN-USB Pro FD channel 1 (device 0)
```

Type command sudo apt-get install can-utils to install can-utils.

Note:

This tool is a very easy way to test USB2CANFD-X2 module communication. There is only a simple use instruction. For more details, please refer to can-utils user manual and source code. https://github.com/linux-can/can-utils/

Send/Receive

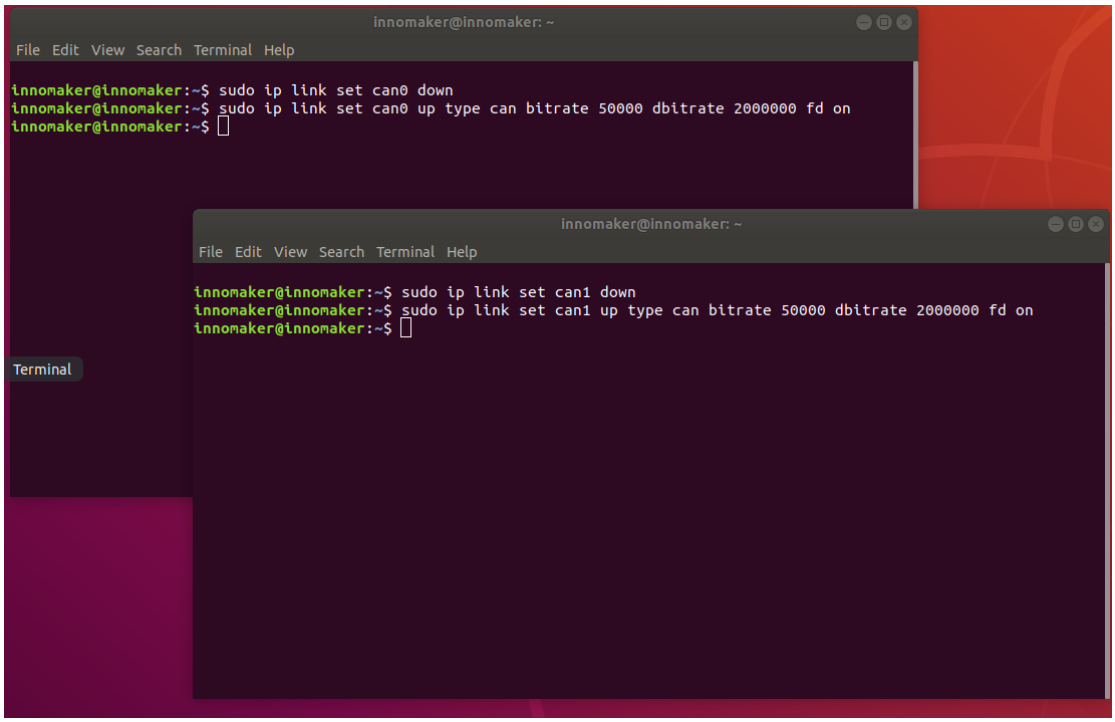
Initialize CAN port, Open two termination command for can0 and can1.

sudo ip link set can0 down

sudo ip link set can0 up type can bitrate 50000 dbitrate 2000000 fd on

sudo ip link set can1 down

sudo ip link set can1 up type can bitrate 50000 dbitrate 2000000 fd on

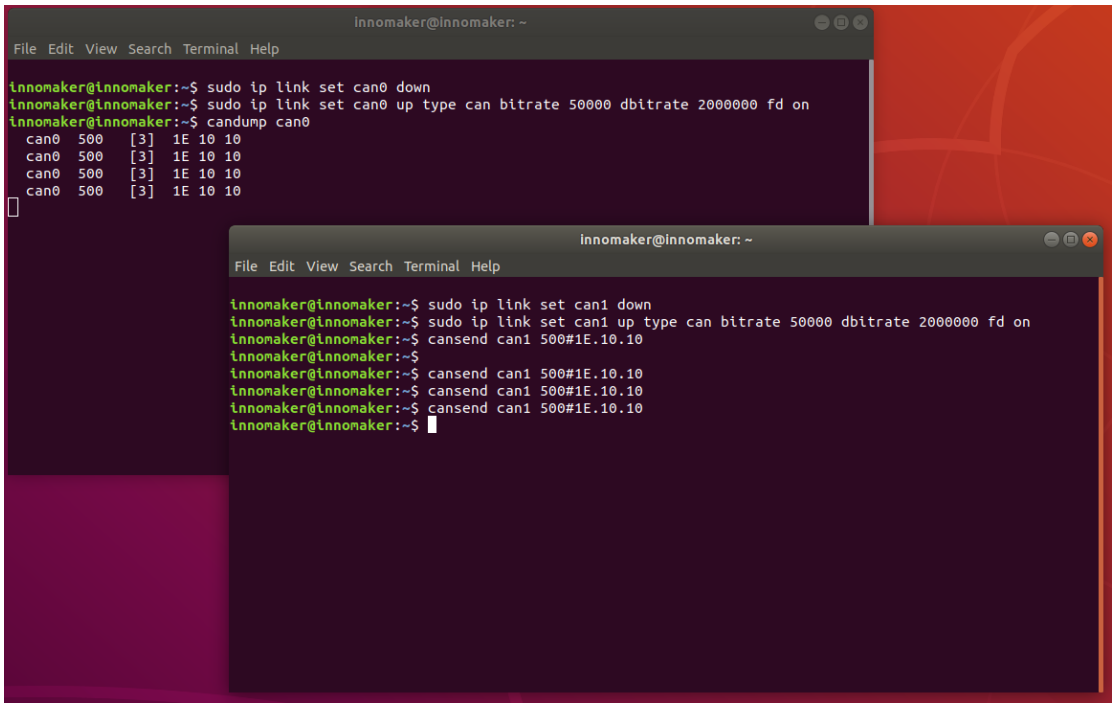


<1>Set can0 as receiver

candump can0

<2>Set can1 as sender

cansend can1 500#1E.10.10



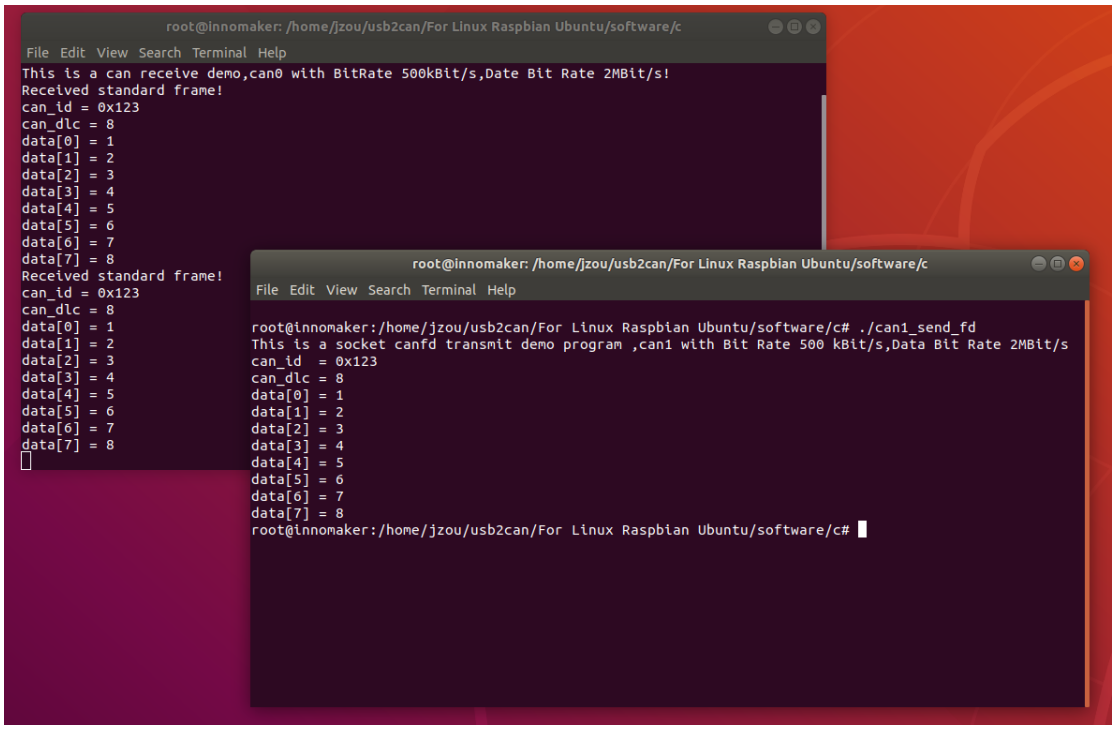
5.6.3 C Demo

<1>Send CAN0 As Receiver,

sudo ./can0_receive_fd

<2>Set CAN1 As Sender

sudo ./can1_send_fd



5.6.4 Python3 Demo

(1) Check the Python version of your Raspbian. Python 3.7.3 default in 2019-09-26-Raspbian.img. Our Demo can run on any Python3 version.

python3 -V

(2) If you can't find the Python3 in system. Install the Python3

sudo apt-get install python3-pip

(3) Install Python CAN library.

sudo pip3 install python-can

(4) Set CAN0 as receiver

```
sudo python3 receive.py
```

(5) Set CAN1 as sender

```
sudo python3 send.py
```

6 TECHNICAL SUPPORT

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